| ditronix.net<br>April 18, 2026 | Board Features | IPEM<br>? | IPEM<br>Plus | EPEM<br>32 | EPEM<br>36 | IPEM<br>PiHat | IPEM<br>PiHat Lite | IPEM<br>32 (EOL) | IPEM<br>36 (EOL) |
|---|---|---|---|---|---|---|---|---|---|
| **Released** | | **2026 May** | **2026 May** | **2025 Dec** | **2025 Dec** | **2025 Apr** | **2025 Apr** | **2023 Feb** | **2023 Feb** |
| **Processing** | Espressif MCU Support | ESP32-C5 | ESP32-C5 | ESP32-C6 | ESP32-C6 | - | - | ESP32 | ESP32 |
| | Raspberry Pi Support (40 Pin) | - | - | - | - | Yes | Yes | - | - |
| **AC Monitoring** | ATM90E32 Energy Monitor Device | Yes | - | Yes | - | - | - | Yes | - |
| | ATM90E36 Energy Monitor Device | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | ATM Fast DMA or SPI Modes | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | ATM90E36 DMA Raw Data Dumping | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | ATM90E36 DMA Data Acquisition | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | Total Harmonic Distortion (THD) | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | Fast Fourier Transform (FFT) | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | Discrete Fourier Transform (DFT) | - | Yes | - | Yes | Yes | Yes | - | Yes |
| | Dynamic Range | 6000:1 | 6000:1 | 6000:1 | 6000:1 | 6000:1 | 6000:1 | 6000:1 | 6000:1 |
| | Active Energy Accuracy | 0.1% | 0.1% | 0.1% | 0.1% | 0.1% | 0.1% | 0.1% | 0.1% |
| | Reactive Energy Accuracy | 0.2% | 0.2% | 0.2% | 0.2% | 0.2% | 0.2% | 0.2% | 0.2% |
| **Configuration** | Board Configuration | Jumper | Jumper | Jumper | Jumper | DIL Switch | DIL Switch | Solder Pad | Solder Pad |
| | Stackable Boards (Extra CT Inputs) | - | - | - | - | - | Yes | - | - |
| **Connections** | Connections 3.5mm Jack (CT) | ? | - | Yes | Yes | Yes | Yes | Yes | Yes |
| | Terminal Blocks | ? | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Current Inputs** | CT Clamp Inputs (Typically 0 to 100A) | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | YHDC SCT013-000 100A / 50mA 1:2000 | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Non YDHC Current Clamp Inputs | Yes | Yes | Yes | Yes | Yes | Yes | - | - |
| | Rogowski Coil Inputs (Typically > 100A) | - | Yes | Yes | Yes | Yes | Yes | - | - |
| | Rogowski Current Transfomer PA3202NL | - | Yes | Yes | Yes | Yes | Yes | - | - |
| | AC Voltage Inputs | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 | 1, 2 or 3 |
| | AC Current Inputs | ? | 4 (3 + N) | 3 | 4 (3 + N) | 4 (3 + N) | 4 (3 + N) | 3 | 4 (3 + N) |
| | Neutral Current Monitoring | - | Yes | - | Yes | Yes | Yes | - | Yes |
| **AC Input** | AC Sense Inputs (Low Voltage) | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| | AC Inputs Configurable (1, 2 or 3) | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Board Power** | AC Power Input SMPS (5 to 20V RMS) | Yes | Yes | - | - | - | - | Yes | Yes |
| | DC Power Input SMPS (5 to 20V) | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | DC USB Power Option | Yes | Yes | Yes | Yes | - | Yes | Yes | Yes |
| **AC Phases** | Single Phase (x3) Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Dual or Split Phase Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Three Phase three-wire (3P3W, Y or Δ) | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Three Phase four-wire (3P4W, Y0) | - | Yes | - | Yes | Yes | Yes | - | Yes |
| **IoT** | Home Assistant Support | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | ESPHome Support | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Tasmota Support | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Domoticz Support | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | MQTT Support | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Connectivity** | LoRa Support | - | Yes | - | - | - | - | - | - |
| | Radio Modules - 2.4 GHz WiFi | Yes | Yes | Yes | Yes | Raspberry Pi | Raspberry Pi | Yes | Yes |
| | Radio Modules - 5 GHz WiFi | Yes | Yes | - | - | Raspberry Pi | Raspberry Pi | - | - |
| | Radio Modules - Bluetooth 2.4 GHz | Yes | Yes | Yes | Yes | Raspberry Pi | Raspberry Pi | Yes | Yes |
| | Radio Modules - Zigbee 2.4 GHz | Yes | Yes | Yes | Yes | - | - | - | - |
| | Radio Modules - Thread 2.4 GHz | Yes | Yes | Yes | Yes | - | - | - | - |
| | Radio Modules - LoRa 850 to 930 MHz | - | Yes | - | - | - | - | - | - |
| | Radio Modules - LoRa SX1262 (FCC/CE) | - | Yes | - | - | - | - | - | - |
| | Ethernet (W5500) | - | - | Yes | Yes | Raspberry Pi | Raspberry Pi | - | - |
| **Interfaces** | Galvanically Isolated Opto Output (50mA) | - | Yes | - | - | - | - | - | - |
| | Realtime Clock Battery Option (CR2032) | - | Yes | Yes | Yes | Yes | - | - | - |
| | OLED Display Interface I2C | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | RS485 Interface | Yes | Yes | Yes | Yes | - | - | - | - |
| | Arduino & PlatformIO Devkit Board | ESP32-C5 | ESP32-C5 | ESP32-C6 | ESP32-C6 | - | - | ESP32 | ESP32 |
| | USB Interface Micro (CH340C UART) | - | - | - | - | - | - | Yes | Yes |
| | USB Interface Type C (CH343P UART) | Yes | Yes | Yes | Yes | - | Power Only | - | - |
| | Expansion Interface Panel I2C | Yes | Yes | - | - | - | - | - | - |
| | Expansion Interface SPI | Yes | Yes | - | - | - | - | - | - |
| | OneWire Support Interface (DS18B20) | Yes | Partly | - | - | - | - | - | - |
| | DAC Output | - | MCU | - | - | MCP4728 | - | - | - |
| | FET Output | ? | Yes | Yes | Yes | - | - | - | - |
| | PWM/GPO Out | ? | Yes | Yes | Yes | - | - | - | - |
| **Other** | MicroSD Card Capabilty | Yes | Yes | - | - | Raspberry Pi | Raspberry Pi | - | - |
| | Internal EEPROM (24C64 64k) | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Realtime Clock DS3231SN | - | Yes | Yes | Yes | Yes | - | - | - |
| | Realtime Clock Battery Backup | - | Yes | Yes | Yes | Yes | - | - | - |
| | ADC Inputs ADS1115 16bit | - | Yes | - | - | Yes | - | - | - |
| | Analogue Temperature Sensor NTC | - | - | - | - | - | - | Yes | Yes |
| | Digital Temperature Sensor TMP102 | Yes | Yes | Yes | Yes | Yes | - | - | - |
| **Mechanical** | DIN Encloure Mount Option | ? | Yes | - | - | - | - | - | - |
| | Display Panel Option | Yes | Yes | - | - | - | - | - | - |
| **Applications** | Home Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Offgrid Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Solar (Farm) Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Energy Gateways | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Data Concentrator | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Electric Metering | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Industrial and Commercial Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| | Farming and Agriculture Monitoring | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

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